Evaluating a Training Policy

Research Group

Department of Economics

Research Seminar

Outline

1. Question and design
2. Results

Road map

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2. Results

Research question

Question

Does training raise participants' earnings?

- Compare eligible and ineligible workers.

Research question

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- Observe earnings before and after implementation.

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Question

Does training raise participants' earnings?

- Compare eligible and ineligible workers.
- Observe earnings before and after implementation.
- Check the plausibility of parallel trends.

Identification

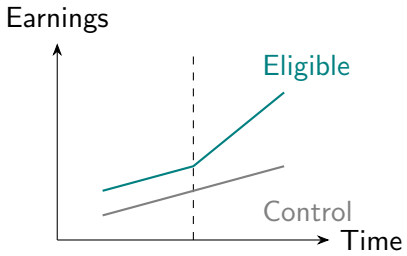
$$Y_{it} = \alpha_i + \lambda_t + \beta(D_i \times Post_t) + \varepsilon_{it}. \tag{1}$$

For an applied discussion, see [1].

Assumption

In the absence of treatment, groups would have parallel trends.

Design in one picture



The dashed line marks implementation. The drawing is schematic and contains no empirical observations.

Road map

1. Question and design
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Illustrative estimates

Table: Demonstration results, not research findings

	Baseline	Controls
Training effect	0.12	0.10
Standard error	0.04	0.03
Observations	800	800

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Report standard errors, units, sample restrictions, and the estimation method.

Interpretation and limitations

- An estimate is meaningful only with a credible design.
- Test alternative windows and inspect pre-treatment trends.
- State the population to which the result applies.

References



Angrist, J. D. and Pischke, J.-S. (2009).
Mostly Harmless Econometrics. Princeton University Press.

Takeaway

Design, evidence, and interpretation should answer the same research question.

Backup: the contrast

$$\widehat{\beta} = (\overline{Y}_{T,post} - \overline{Y}_{T,pre}) - (\overline{Y}_{C,post} - \overline{Y}_{C,pre}).$$